



The Hashemite University

Faculty of Prince Al Hussein Bin Abdallah II for Information Technology

Department of Software Engineering

Software Testing

Project 2: Formal Criteria-Based Test Design & Automation

1. ISP Design

1.1 Characteristics

Characteristic	Description
customerType	Represents the type of customer
totalOrdersInLastYear	Represents the number of orders placed in the last year
isSubscribedToNewsletter	Indicates whether the customer is subscribed to the newsletter

1.2 Input Space Partitions

customerType Blocks

Block ID	Values
CT1	NEW
CT2	REGULAR
CT3	PREMIUM

totalOrdersInLastYear Blocks

Block ID	Values
TO1	totalOrdersInLastYear < 10
TO2	totalOrdersInLastYear >= 10

isSubscribedToNewsletter Blocks

Block ID	Values
NS1	true
NS2	false

The input characteristics were divided into disjoint and complete partitions. Each block represents a unique system behavior, and together the partitions cover all possible valid inputs.

2. Pair-Wise Test Cases

Test Case	customerType	totalOrdersInLastYear	isSubscribedToNewsletter	Expected Discount
TC1	NEW	5	false	5%
TC2	NEW	5	true	7%
TC3	REGULAR	5	false	8%
TC4	REGULAR	10	true	15%
TC5	PREMIUM	5	true	12%
TC6	PREMIUM	10	false	15%

These test cases were designed using Pair-Wise Coverage (PWC) to ensure that every pair of input values appears together in at least one test case while minimizing the total number of tests.

3. Infeasible Combination

customerType	totalOrdersInLastYear
NEW	>= 10

A NEW customer cannot logically have 10 or more previous orders because a new customer has not yet established purchase history. Therefore, this combination is considered infeasible and the system should throw an `IllegalArgumentException`.

4. Limitations of Pair-Wise Testing

Pair-Wise Testing ensures that every pair of input values is tested at least once. However, achieving 100% Pair-Wise Coverage does not guarantee a completely bug-free system. Some defects may only appear due to interactions between three or more input variables simultaneously. As a result, Pair-Wise Testing is more efficient than exhaustive testing in reducing the number of test cases, but it may still miss certain complex interaction faults.

5. Team Contribution Matrix

Member	Contribution
Amer Shdifat (2233532)	Designed the ISP analysis, and implement the DiscountCalculator class
Abdallah Abu Olaim (2235814)	Implement the JUnit parameterized tests and exception testing

Thank You